Life Expectancy Prediction

BUS235C

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Abstract

Life expectancy is a crucial metric describing the overall well-being and development of a population. It plays a pivotal role in helping governments understand the effect of public health policies and in allocating funds for retirement programs. In this work, we have explored the relationship between life expectancy and other socio-economic and health metrics of a population such as mortality rates, immunization coverage, and education. We have predicted the life expectancy for a range of countries for the year 2015, given historical data for the years 2000-2014. We found that linear regression is a good model to make predictions for this dataset. However, we improved upon the performance of linear regression by adding data regarding a country's region in the form of dummy variables. We have also explored the application of other methods such as regression trees, neural networks, and boosting methods such as XGBoost.

1. Introduction

Life expectancy at birth is a statistical measure of the average expected lifespan at the time of birth within a population. It is a crucial metric for understanding the overall health, well-being, and development of a population.

Generally, a higher life expectancy value for a country suggests better healthcare access, disease prevention and general living conditions. A lower life expectancy can conversely be caused due to poor access to healthcare, widespread diseases and higher mortality rates. Apart from the direct correlation with public health, life expectancy is also closely linked to the level of economic development of a country. Wealthy nations can invest more in healthcare and infrastructure while poorer nations with limited resources will struggle with challenges in nutrition and sanitation. There are several other factors which can affect life expectancy directly or indirectly. Health habits such as alcoholism and smoking, environmental factors such as air pollution, immunization rates, food habits, obesity and education all affect life expectancy of a country to various degrees.

Life expectancy is a key metric for evaluating the effectiveness of implementing public health policies at scale. It is also an essential metric for helping governments plan old-age support programs such as U.S. Social Security and pension, since the financial strategy to support such programs is heavily dependent on the expected lifespan of a country's population.

In this project, we aim to explore the complex relationships between life expectancy and factors such as economics, health and mortality, society and life choices. Our goal is two-fold – first, to be able to predict life expectancy of a given population provided information regarding other factors; and second, to identify the factors most strongly contributing to low life expectancies in a given country.

To achieve these goals, we have applied several techniques including linear regression, regression trees, neural networks and boosting methods to perform life expectancy predictions. We have also performed variable selection to identify the smallest model with good prediction accuracy on the training dataset that incorporates only the most impactful variables. We have uncovered several important correlations within the dataset and identified the strengths and shortcomings of various models. We are able to achieve very good accuracy on predictions over the test dataset while improving our understanding of the data.

2. Data Description

The dataset has been obtained from Kaggle (link) and contains life expectancy, health, mortality, immunization, economic and demographic information about 179 countries between the years 2000 and 2015. The dataset has 20 independent variables, 1 dependent variable (life expectancy) and 2864 rows.

The data has been collated from various official sources such as World Bank, World Health Organization (WHO) and the University of Oxford. Detailed references for each subset of data have been provided in Section 6. Detailed description of each column is provided below.

Column Name	**Description**
Country	Country
Region	Geographical region
Year	Year between 2000 and 2015
Infant_deaths	Number of infant deaths per 1000 population
Under_five_deaths	Number of deaths of children under 5 per 1000 population
Adult_mortality	Adult mortality rates of both sexes per 1000 population
Alcohol_consumption	Per capita alcohol consumption in litres
Hepatitis_B	Hepatitis_B immunization coverage among 1 year olds (%)
Measles	Measles immunization coverage among 1 year olds (%)
BMI	Body Mass Index
Polio	Polio immunization coverage among 1 year olds (%)
Diphtheria	Diphtheria immunization coverage among 1 year olds (%)
Incidents_HIV	Incidents of HIV per 1000 poputlation aged 15-49
GDP_per_capita	Per capita gross domestic product
Population_mln	Total population in millions
Thinness_ten_nineteen_years	Prevalence of thinness among adoloscents aged 10-19 years
Thinness_five_nine_years	Prevalence of thinness among children aged 5-9 years
Schooling	Average years people aged 25+ spent in formal education
Economic_status_Developed	1 if country is categorized as ‘Developed’
Economic_status_Developing	1 if country is categorized as ‘Developing’
Life_expectancy	Life expectancy of population in years

Table 1: Description of dataset columns

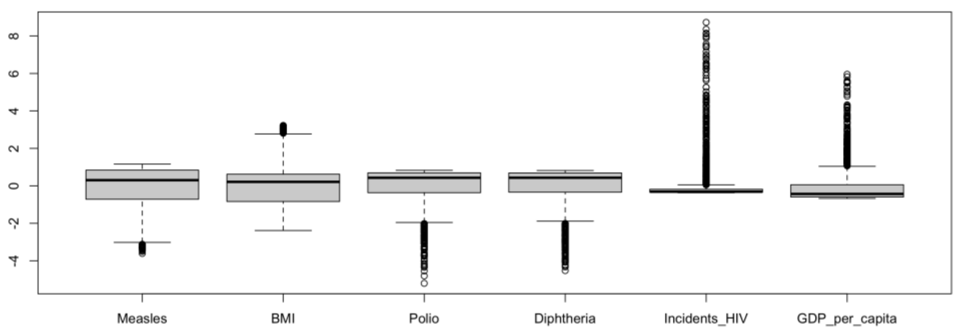
Amongst the 20 independent variables, there are two categorical variables – Country and Region. Also, there are two dummy variables to indicate the economic status of a country as ‘Developed’ or ‘Developing’.

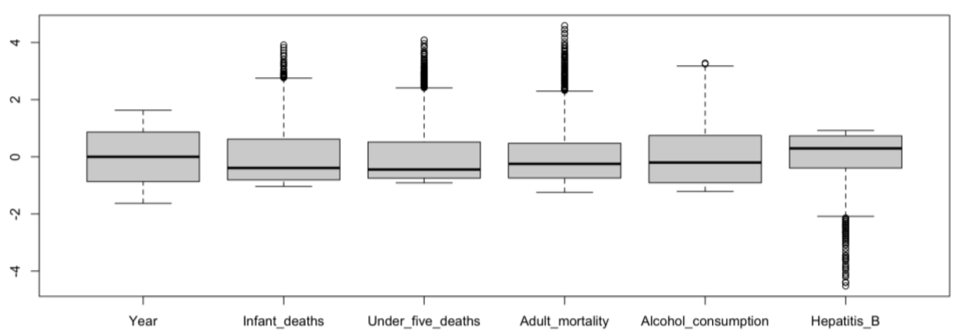
3. Exploratory Data Analysis

The summary statistics of the dataset reveal some interesting facts:

- Nearly 20% of the countries are categorized as ‘Developed’ and 80% as developing.
- The life expectancy values range from a minimum of 39.4 years to a maximum of 83.8 years with a mean value of 68.86 years.

In order to identify outliers in the data, we have plotted the boxplot of all independent variables after scaling.





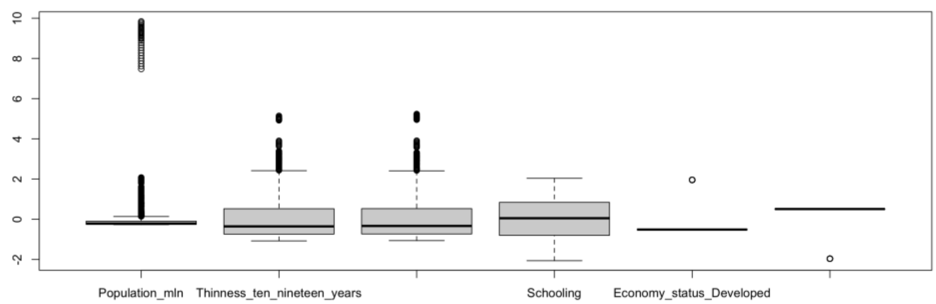


Figure 1: Box plot of all numerical independent variables in the training dataset after scaling.

We can see from Figure 1 that all independent variables have outliers (except Year since it is regularly spaced between 2000 and 2015). In particular we notice that 'Incidents_HIV', 'population_mln' and 'GDP_per_capita' have a large number of outliers with a big range. To investigate this further, we look at the distribution of their values using histograms (Figure 2).

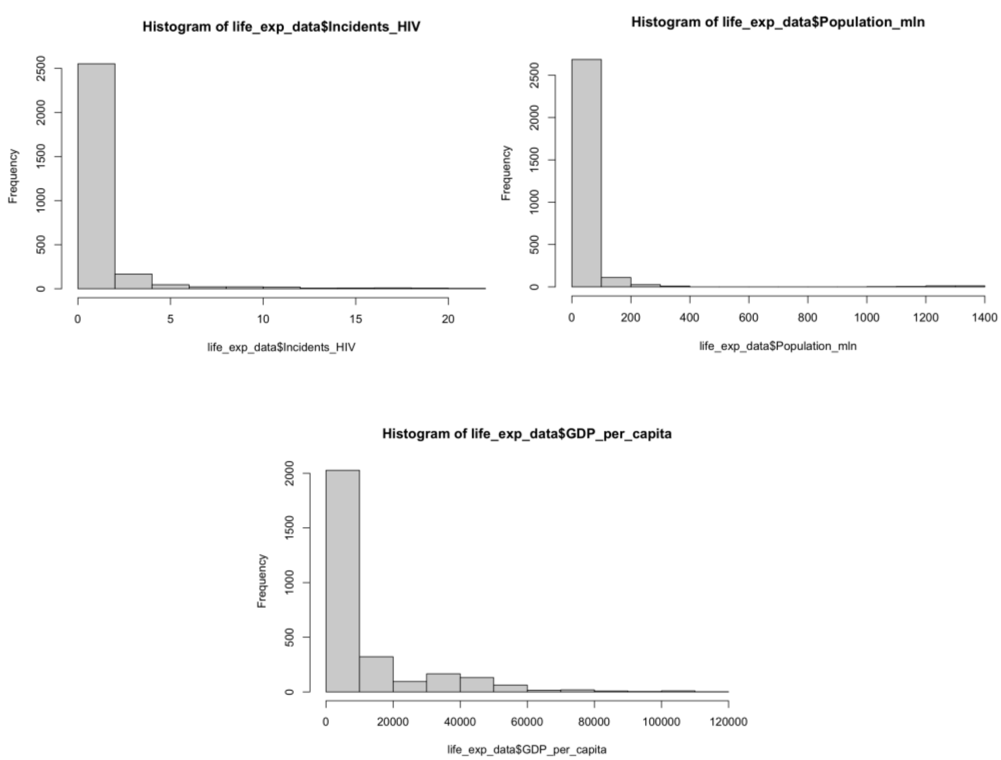


Figure 2: Histograms of variables with a large number of outliers

From Figure 2, we can see that indeed the distributions for these variables are severely skewed with most countries falling in the lowest value bucket. Next, we wanted to explore any correlations that exist within the dataset by plotting a heatmap of the correlations for each pair of variables as depicted in Figure 3. From Figure 3, we can make the following observations:

- A strong negative correlation between mortality variables (infant_deaths, under_five_deaths, adult_mortality) and life_expectancy. This is expected since life expectancy at birth is directly impacted by these metrics.
- If a country is categorized as 'Developed', the life expectancy is correlated positively and is expected to be higher than in developing countries.
- A moderate positive correlation exists between schooling and life_expectancy. This is an interesting observation as there is no direct relationship between these two variables.
- We can also see that higher rate of immunization lead to higher life expectancy.

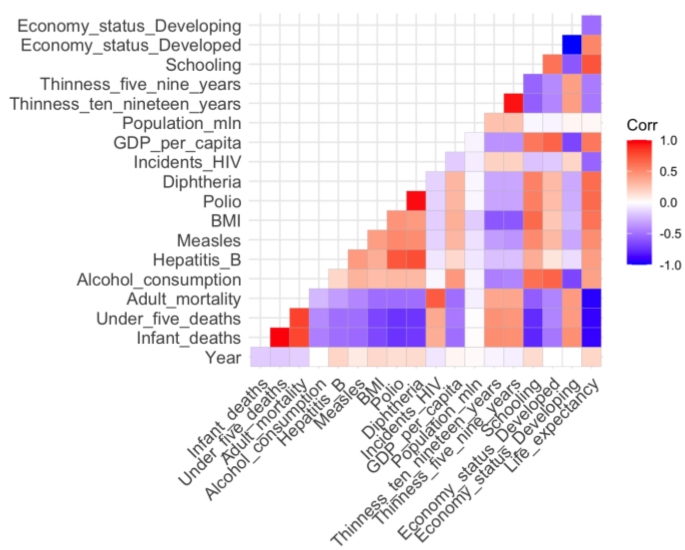


Figure 3: Heatmap of correlations between the independent variables of the dataset

Next, we can look at how some variables are distributed by region. Figure 4 (a) shows the scatter plot of per capita GDP of a country vs the life expectancy, with different colors for different regions. We can see that for every region of the world, life expectancy tends to increase with GDP really quickly in the beginning (e.g., in Africa and South America). However, very high values of per capita GDP do not add too much to the life expectancy (e.g., European Union).

We have also plotted Schooling vs life expectancy by region in Figure 4(b). We note that for most regions, there is a definite positive correlation between life expectancy and schooling (large number of average years of formal education). However, the same is not observed as consistently for some other regions such as Asia and Africa.

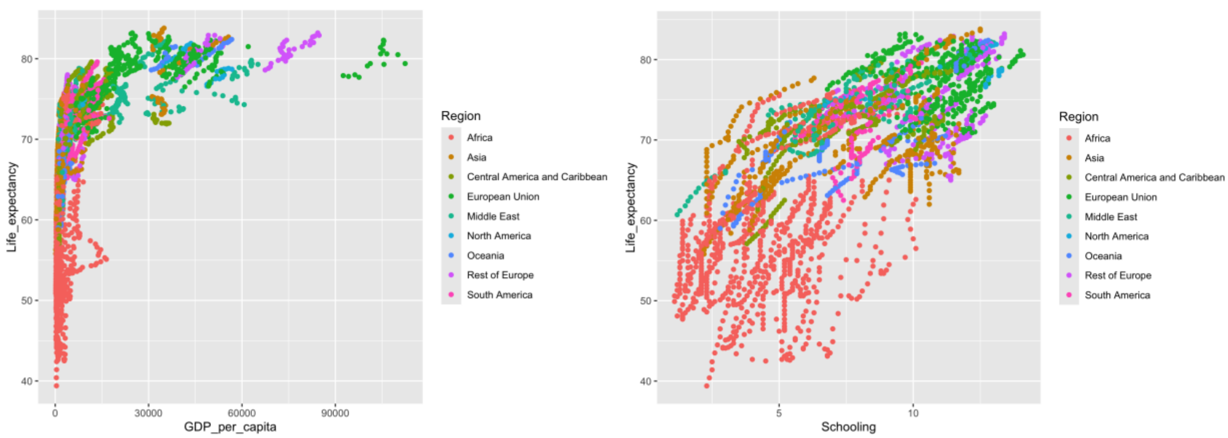


Figure 4: (a) Scatter plot of Life expectancy vs per capita GDP by region and (b) Scatter plot of Life expectancy vs Schooling by region

In order to do some preliminary analysis, we fit a linear regression model to the dataset. First, we dropped the columns of ‘Country’ and ‘Region’ as they may lead to some countries not showing up in the test dataset when randomly sampled. Then, we divided the data into 80% training and 20% test. We fit the linear regression model with ‘Life_expectancy’ as the dependent variable and the other 18 variables as independent. We obtained an adjusted R^2 value of 0.9793 on the training dataset. The MSE value on the training dataset was 1.8287 and on the test dataset was 1.8426. Clearly, this model is well-suited for regression-based methods.

4. Results

For a variable such as life expectancy, it makes most sense to predict data temporally, as it is most practical to predict future life expectancy, based on historical data. Hence, we have set the training data to be the yearly life expectancy data from 2000-2014 for 179 countries. The test dataset contains the life expectancy data for the same set of 179 countries for the year 2015. This gives us 2685 rows in the training data and 179 rows in the testing dataset. To begin with, we drop all categorical variables (Country and region) from the dataset and fit a linear regression model using remaining numerical variables. We obtain an R^2 value of 0.9795 and in-sample and out-of-sample MSE of 1.823 and 1.932 respectively (Table 2 details the performance of all models tested in this project). This is a very respectable performance, and the model is already providing good results.

Model	Details	Adjusted R^2	In-sample MSE	Out-of-sample MSE
Linear Regression	No country/region info	0.9795	1.8234	1.932
Linear Regression	Including dummy variables for *Region*	0.9841	1.4127	1.4946
Linear Regression	Including dummy variables for *Region* Variable selection using AIC	0.9841	1.4142	1.5021
Linear Regression	Including dummy variables for *Region* Variable selection using BIC	0.9840	1.4283	1.5105
Regression Tree	Including dummy variables for *Region*	NA	6.2662	7.3424
Neural Network	1 hidden layer, 3 neurons	NA	0.8102	10.7218
Neural Network	1 hidden layer, 5 neurons	NA	0.5742	12.7334
XGBoost	100 rounds, squared error loss	NA	0.0121	0.3631

Table 2: Performance of all models tested with the life expectancy dataset.

However, we noted that the prediction accuracy of some regions is worse than others as shown by the breakdown of MSE by region in Fig. 5.

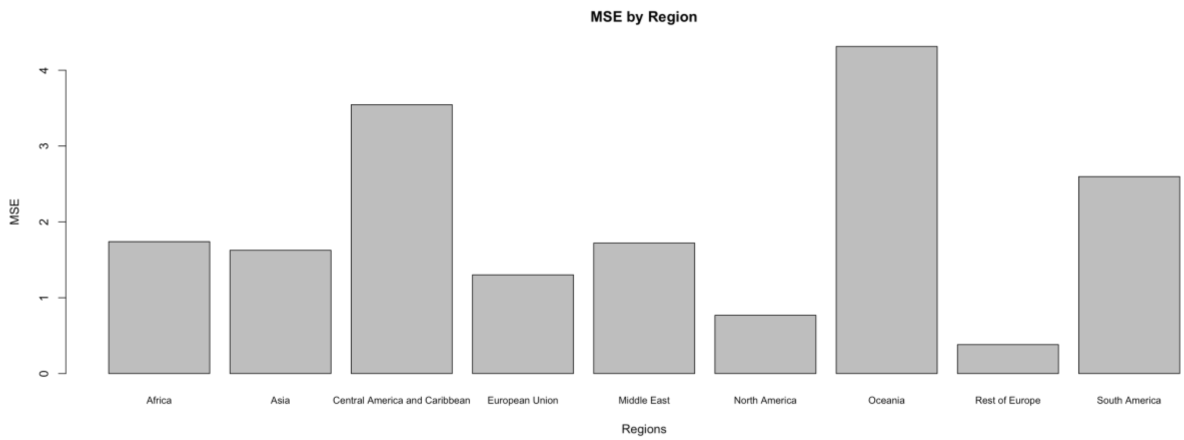


Figure 5: Mean Squared Error breakdown by region for a model trained without region information.

We can see that the MSE is much higher for some regions such as Central America and Caribbean, Oceania, and South America, compared to other regions of the world. We believe this is due to the underrepresentation of these regions in the test dataset due to a smaller number of countries that belong to these regions, and also because they have a unique pattern in their life expectancy data which a general model is not able to capture accurately.

To account for this problem, we added 8 dummy variables to our dataset to represent the 9 regions, bringing the total number of independent variables in the model to 26. Performing linear regression with this model, we achieve a higher adjusted R^2 value of 0.9841, and a lower in-sample and out-of-sample MSE of 1.4127 and 1.4946. Taking a look at the breakdown of MSE by region after adding dummy variables (Fig. 6), we can see that MSE for the Central America and Caribbean, Oceania, and South America has improved significantly.

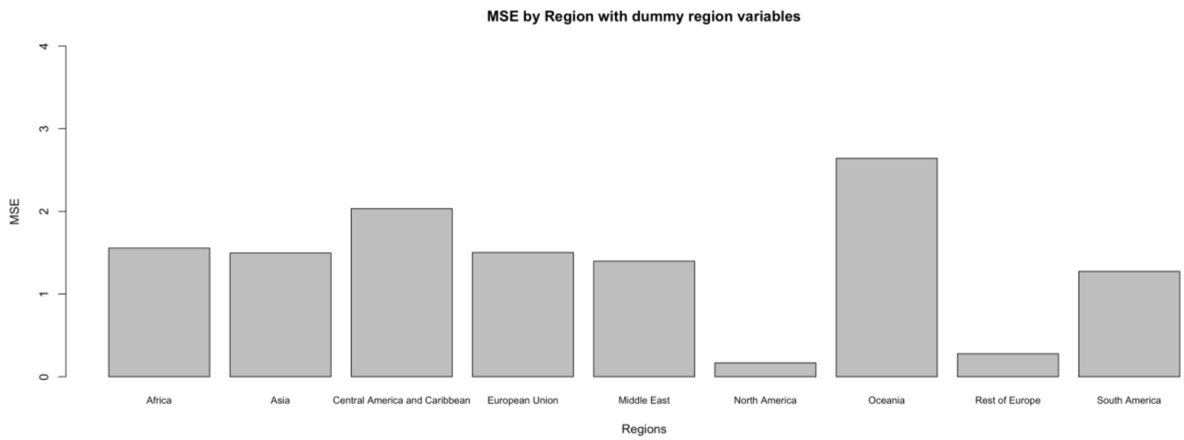


Figure 6: Improvement in Mean Squared Error after incorporating region information in the model.

Next, we wanted to identify the most important independent variables which affect life expectancy in a population, while reducing the size of the models. To do this, we performed variable selection using backward elimination starting with all independent variables. By minimizing AIC, we achieved a model containing 22 variables which eliminated Alcohol_consumption, Measles, Population_mln, Economy_status_Developing. The elimination of Measles and Economy_status_Developing makes sense since the information contained in them is already well represented by other variables that have strong correlations with these variables, and they do not add much new information. It is interesting to note however, that the population of a country and the alcohol consumption rate are amongst the variables that are eliminated from the model. Selecting a model via BIC results in a model containing 14 variables, with 12 variables dropped. The selected variables include RegionCentral.America.and.Carribean, RegionEuropean.Union, RegionOceania, RegionSouth.America, Year, Infant_deaths, Under_five_deaths, Adult_mortality, Hepatitis_B, BMI, Incidents_HIV, GDP_per_capita, Schooling, and Economy_status_Developed. This list of variables is indicative of the most important variables when it comes to life expectancy prediction.

Next, we wanted to explore other data mining methods that can provide good performance for life expectancy prediction. We trained a regression tree model using all independent variables which resulted in an in-sample MSE of 6.2662 and out-of-sample MSE of 7.3424. These error values are clearly significantly higher compared to linear regression. The reason behind high error values becomes clear when we look at the structure of the tree that was learnt by the model (Figure 7).

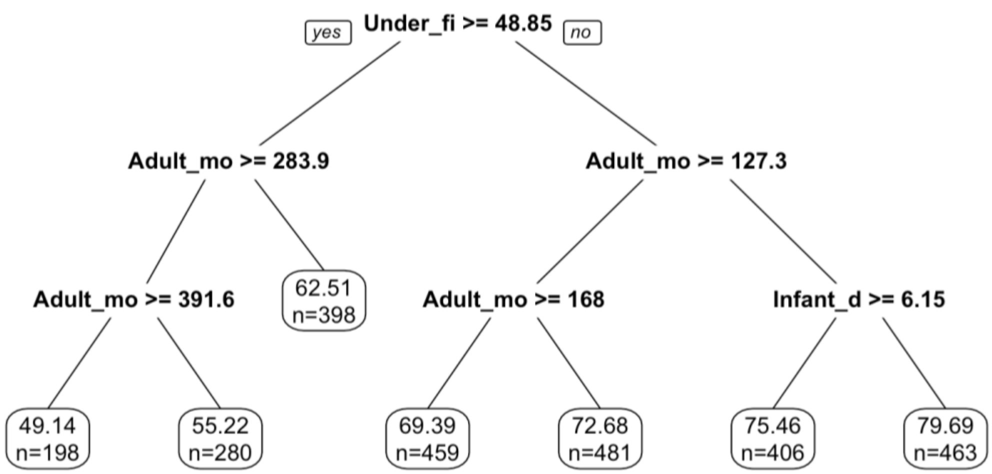


Figure 7: Regression tree model for life expectancy trained on data from 2000 to 2014 for 179 countries.

We note that the only variables being used by the model to make predictions are the mortality rates (Under_five_deaths, infant_deaths, and adult_mortality). The model is very simplistic and is not able to capture the details of patterns within the training dataset and the interdependence of other independent variables. Hence the performance of regression tree on this dataset is not great.

We also trained two neural network models with a single hidden layer and 3 and 5 neurons in the hidden layer respectively. With the 3-neuron model, the in-sample MSE is 0.8102 and the out-of-sample MSE is 10.7218. As we can see, the in-sample MSE is quite low, but the out-of-sample MSE is significantly higher which indicates that the model is overfitting. The large number of parameters in a neural network model leads to the model learning the noise in the data as patterns, hence, the out-of-sample performance is much worse. Increasing the number of neurons in the hidden layer to 5 leads to an even lower in-sample MSE of 0.5742 alongside an even larger out-of-sample MSE of 12.7334 which leads credence to our theory that these neural network models are overfitting on the training dataset.

Finally, we explored a new model called XGBoost [14] to perform life expectancy predictions. XGBoost is a particular variation of the machine learning method called 'Boosting'. The idea behind boosting methods is that a large number of weak learners, i.e., very simple models which are not very accurate by themselves, can provide very good prediction accuracy when combined together in an ensemble to create a strong learner. Gradient boosting is a variation of boosting wherein each new model is trained to minimize the errors of the previous model using gradient descent. XGBoost (Xtreme Gradient Boost) in particular, is a further variation of gradient boosting that has been optimized to handle large datasets, efficiently handle missing values, and has built-in support for parallel processing. These features have made XGBoost a popular choice for providing very good performance on several machine learning tasks. In this project, we used XGBoost by specifying squared error as the cost function and performing 100 rounds of boosting using an ensemble of simple decision trees to create a strong model. When using XGBoost, the in-sample MSE and out-of-sample MSE are 0.0121 and 0.3631 respectively, which are significantly better than any other methods tested on this dataset.

5. Conclusions

In this project, we have trained several models on historical data of life expectancy, socio-economic and health conditions for 179 countries spanning 15 years, to perform prediction of life expectancy for the subsequent year. We have shown that linear regression provides very good performance on this dataset, and that accuracy can be improved when data regarding a country's region is included in the model. Furthermore, we found that regression tree models are overly simplistic for this dataset, and neural network models are overly complex. While we see large errors for regression trees, neural network models can easily overfit on the training data and provide poor performance on the test dataset. Boosting methods such as XGBoost provide the perfect balance for this dataset and result in very low errors on the test dataset.

6. References

Detailed references for sources of data are provided below (as listed on the data source link):

1. Average life expectancy of both genders in different years from 2010 to 2015 : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/life-expectancy-at-birth-(years)
2. Mortality-related attributes (infant deaths, under-five-deaths, adult mortality) : https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates

3. Alcohol consumption that is recorded in liters of pure alcohol per capita with 15+ years old : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/alcohol-recorded-per-capita-(15-)-consumption-(in-litres-of-pure-alcohol)
4. % of coverage of Hepatitis B (HepB3) immunization among 1-year-olds : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/hepatitis-b-(hepb3)-immunization-coverage-among-1-year-olds-(-)
5. % of coverage of Measles containing vaccine first dose (MCV1) immunization among 1-year-olds : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/measles-containing-vaccine-first-dose-(mcv1)-immunization-coverage-among-1-year-olds-(-)
6. % of coverage of Polio (Pol3) immunization among 1-year-olds : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/polio-(pol3)-immunization-coverage-among-1-year-olds-(-)
7. % of coverage of Diphtheria tetanus toxoid and pertussis (DTP3) immunization among 1-year-olds : https://www.who.int/data/gho/data/indicators/indicator-details/GHO/diphtheria-tetanus-toxoid-and-pertussis-(dtp3)-immunization-coverage-among-1-year-olds-(-)
8. BMI : https://www.who.int/europe/news-room/fact-sheets/item/a-healthy-lifestyle---who-recommendations
9. Incidents of HIV per 1000 population aged 15-49: https://data.worldbank.org/indicator/SH.HIV.INCD.ZS
10. Prevalence of thinness among adolescents aged 10-19 years. BMI < -2 standard deviations below the median: https://www.who.int/data/gho/indicator-metadata-registry/imr-details/4805

11. GDP per capita in current USD: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?most_recent_year_desc=true
12. Total population in millions: https://data.worldbank.org/indicator/SP.POP.TOTL?most_recent_year_desc=true
13. Average years that people aged 25+ spent in formal education: https://ourworldindata.org/grapher/mean-years-of-schooling-long-run
14. XGBoost: https://dl.acm.org/doi/abs/10.1145/2939672.2939785

Chen, T., & Guestrin, C. (2016, August). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785-794)